

Name : _____

Date : _____ 20

Examination : _____ Branch/Semester _____

Subject : _____

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AML (Honours - LY)

Q1:- consider a simple linear regression problem with single feature (x) and target (y). Start with initial values of m (slope) and c (intercept) as 0.

The eqⁿ of model is $y = mx + c$

→ consider dataset

x	y
1	3
2	4
3	7
4	8

The learning rate is 0.1. answer following

- calculate initial prediction and error for each instance in the dataset.
- compute gradients of loss with respect to m and c .
- Update the values of m and c using gradients computed in the previous step and learning rate.

⇒ Solution

a) Initial Prediction and error for each instance.

Given $m = 0$ and $c = 0$.

$$\hat{y}_i = mx_i + c = 0$$

∴ Errors are

→ for $(x_1, y_1) = (1, 3)$, $\hat{y}_1 = 0$.

Error₁ = $\hat{y}_1 - y_1 = 0 - 3 = -3$.

→ for $(x_2, y_2) = (2, 4)$, $\hat{y}_2 = 0$ ∴ error₂ = $0 - 4 = -4$.

→ Similar. for (x_3, y_3) .

$$\text{Error}_3 = 0 - 7 = \underline{-7}$$

→ for (x_4, y_4)

$$\text{Error}_4 = 0 - 8 = \underline{-8}$$

b) Gradients of loss with respect to m and c .

→ The Mean Squared Error

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \rightarrow \text{Loss.}$$

∴ Gradients w.r. to m .

$$\frac{\partial \text{MSE}}{\partial m} = \frac{2}{n} \sum_{i=1}^n (mx_i + c - y_i) \cdot x_i$$

→ Sub $m, c = 0$.

$$\frac{\partial \text{MSE}}{\partial m} = \frac{2}{4} [-3 \times 1 + (-4) \times 2 + (-7) \times 3 + (-8) \times 4]$$

$$= \frac{2}{4} [-3 - 8 - 21 - 32]$$

$$= \frac{2}{4} [-64] = \underline{\underline{-32}}$$

→ Similar. with respect to c .

$$\frac{\partial \text{MSE}}{\partial c} = \frac{2}{n} \sum_{i=1}^n (mx_i + c - y_i)$$

→ Put m and $c = 0$.

$$\therefore \frac{2}{4} [-3 - 4 - 7 - 8] = \frac{2}{4} [-22] = \underline{\underline{-11}}$$

c) Update the values of m and c using the gradients and learning rate.

$$m_{\text{new}} = m_{\text{old}} - \alpha \frac{\partial \text{MSE}}{\partial m}$$

$$2.0 = 2.0 + 0 = (2.0) \quad \frac{\partial m}{\partial m} = 1$$

$$2.0 = 2.0 + 0.1 \times (-32) = 2.0 - 3.2 = -1.2$$

$$\rightarrow c_{\text{new}} = c_{\text{old}} - \alpha \frac{\partial \text{MSE}}{\partial c}$$

$$0.0 = 0.0 + 0.1 \times (-11) = 0.0 - 1.1 = -1.1$$

$\therefore$ Updated values of m and c after one iteration are.

$$\boxed{\begin{matrix} m = -1.2 \\ c = -1.1 \end{matrix}}$$

Note $\rightarrow$ Now these updated values will be used for next iteration.

* Now Same Problem for SGD i.e. Stochastic Gradient Descent and Mini Batch is.

$\rightarrow$ In SGD, after each individual data point, coefficients are updated.

for e.g. for $(x, y) = (1, 3)$ {initially m and $c = 0$ }.

$$\text{Error} = \hat{y}_i - y_i = 0 - 3 = -3$$

$$\text{Gradient w.r.to } m = \frac{\partial \text{MSE}}{\partial m} = \frac{2}{n} \cdot (\hat{y}_i - y_i) \cdot x_i$$

$$\text{and } \frac{\partial \text{MSE}}{\partial c} = \frac{2}{1} \times (-3) = -6$$

$\therefore$ new value of m and c are.

$$m_{\text{new}} = 0 - 0.1(-6) = 0 + 0.6 = 0.6$$

$$c_{\text{new}} = 0 - 0.1(-6) = 0 + 0.6 = 0.6$$

→ Now for $(x_2, y_2) = (2, 4)$

$$\text{Prediction } \hat{y}_2 = 0.6 \times 2 + 0.6 = 1.8$$

$$\text{Error} = \hat{y}_2 - y_2 = 1.8 - 4 = -2.2$$

$$\therefore \frac{\partial \text{MSE}}{\partial m} = \frac{2}{n} \cdot (-2.2) \cdot 2 = -8.8$$

$$\frac{\partial \text{MSE}}{\partial c} = \frac{2}{n} \cdot (-2.2) = \frac{2}{1} \cdot (-2.2) = -4.4$$

$$\therefore m_{\text{new}} = 0.6 - 0.1 \cdot (-8.8) = 0.6 + 1.88 = 1.48$$

$$c_{\text{new}} = 0.6 - 0.1 \cdot (-4.4) = 1.04$$

→ Similarly for (x_3, y_3) and (x_4, y_4) .

* For Minibatch (Batch size 2)

↳ consider Batch 1. (x_1, y_1) and (x_2, y_2) .

and Batch 2 = (x_3, y_3) and (x_4, y_4) .

⇒ For $(x_1, y_1) = (1, 3)$

$$\text{Prediction } \hat{y}_1 = 0.1 + 0 = 0.1$$

$$\text{Error } \hat{y}_1 - y_1 = 0.1 - 3 = -2.9$$

$$\text{and gradient } \frac{\partial \text{MSE}_1}{\partial m} = \frac{2}{2} \cdot (-3) \cdot 1 = -3 \quad \left\{ \text{i.e. } \frac{2}{N} (\text{error}) \cdot (x) \right\}$$

$$\frac{\partial \text{MSE}_1}{\partial c} = \frac{2}{2} \cdot (-3) = -3$$

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Similarly for $(x_2, y_2) = (2, 4)$

$$\text{Prediction } \hat{y}_2 = 0.2 + 0 = 0$$

$$\text{Error } \hat{y}_2 - y_2 = -4$$

$$\frac{\partial \text{MSE}_2}{\partial m} = \frac{2}{2} (-4) \cdot 2 = -8$$

$$\frac{\partial \text{MSE}_2}{\partial c} = \frac{2}{2} (-4) = -4$$

→ Average gradients of Batch 1

$$\frac{\partial \text{MSE}_{\text{batch1}}}{\partial m} = \frac{-3 + (-8)}{2} = -5.5$$

$$\frac{\partial \text{MSE}_{\text{batch1}}}{\partial c} = \frac{-3 + (-4)}{2} = -3.5$$

∴ update m and c after batch 1

$$m_{\text{new}} = m - \alpha \frac{\partial \text{MSE}_{\text{batch1}}}{\partial m} = 0 - 0.1(-5.5) = 0.55$$

$$c_{\text{new}} = 0 - 0.1(-3.5) = 0.35$$

→ Similarly calculate m_{new} and c_{new} for Batch 2. in batch 2 $(x_3, y_3) = (3, 7)$ and $(x_4, y_4) = (4, 8)$.

Momentum Based Gradient descent

It adds a momentum term to the Gradient update to help accelerate the convergence and smooth out the updates. The momentum term accumulates the past gradients to determine the direction and speed of updates.

→ Mean square error $y = \text{loss} = L$.

→ Given α and momentum coefficient (β).

→ γ_m and γ_c are momentum for slope

→ Initially γ_m and $\gamma_c = 0$ and intercept.

$$\gamma_m = \beta \gamma_m + \alpha \frac{\partial L}{\partial m}$$

$$\text{and } \gamma_c = \beta \gamma_c + \alpha \frac{\partial L}{\partial c}$$

→ update coefficients.

$$m_{\text{new}} = m - \gamma_m$$

$$c_{\text{new}} = c - \gamma_c$$

$$m_{\text{new}} = (2.7 - 1.0) = 1.7$$

update coefficients after 1st iteration:

$$m_{\text{new}} = (2.7 - 1.0) = 1.7$$

$$c_{\text{new}} = (2.8 - 0.1) = 2.7$$